

# Installing Python

## A small guide using uv

### Why we use uv for this course

uv is a new (and very fast) Python tool written in Rust. It:

- Installs Python for you (no manual downloads).
- Creates isolated *virtual environments* (safe sandboxes per project).
- Installs and updates packages quickly.

#### Note

WHAT is a virtual environment? Think of each project as its own coffee shop with its own supplies. One shop changing its menu does not affect the others. WHY it matters: You avoid random breakage when different projects need different versions of the same package.

### Install uv

Choose the instructions for your operating system. To these, you need to work in your terminal.

#### Tip

Open the terminal in VS Code by selecting (Terminal → New Terminal).

### macOS or Linux (Terminal)

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

If curl is missing:

```
wget -qO- https://astral.sh/uv/install.sh | sh
```

After installation: close and reopen your terminal (so your PATH updates).

### Windows (PowerShell)

Open PowerShell and run:

```
powershell -ExecutionPolicy Bypass -c "irm https://astral.sh/uv/install.ps1  
| iex"
```

If you see a script execution warning, you can alternatively first run:

```
Set-ExecutionPolicy -ExecutionPolicy Bypass -Scope Process
```

Then re-run the install line.

## Verify installation

Run (macOS / Linux / Windows):

```
uv --version
```

If you see a version number: great!

### Warning

If you get “command not found” or “uv’ is not recognized”:

1. Close and reopen the terminal (important).
2. On Windows: make sure you used PowerShell (not Command Prompt).
3. Still broken? Ask for help, no need of guessing the error.

## InstallPython

We want everyone on the same Python version for consistency. Thus, we’ll use Python 3.12 for the course this year.

Install (you only need to do this once):

```
uv python install 3.12
```

Check the installation:

```
uv run python --version
```

Expected output starts with:

```
Python 3.12.
```

## Create your first project

Pick a folder where you keep course work. If you do not have one, make sure to create one! Open the course folder in your IDE and then run the following from the terminal:

```
uv init
```

`uv init` creates:

- `main.py` (starter script)

- `pyproject.toml` (project + dependencies config)
- `.python-version` (records the Python version we chose)
- `.gitignore` (useful if you ever use Git)
- `README.md` (you can jot notes here)
- (A `.venv` folder will appear later once packages are added or synced.)

#### Note

You do not need to edit any of these (except maybe `README.md` for your notes and `main.py` if you want to run something different).

## Run the starter script

Inside the project folder in your terminal:

```
uv run python main.py
```

You should see something like:

```
Hello World!
```

(If you want, you can open `main.py` and change the message, then re-run.)

## What does that code mean?

```
def main():
    print("Hello, World!")

if __name__ == "__main__":
    main()
```

- `def main():` defines a function (a reusable block of code).
- `print(...)` shows text in the terminal.
- The line `if __name__ == "__main__":` ensures this only auto-runs when the file is executed directly.

Don't worry about this yet, we'll gradually build up to it.

## Adding packages (later in the course)

If/when you need a package (example: `pandas`):

```
uv add pandas
```

If you added the wrong one:

```
uv remove pandas
```

If your `pyproject.toml` changed (e.g. you pulled code from someone else):

```
uv sync
```

#### Tip

If something seems “off”, just close the terminal and reopen in the project folder. Fresh starts fix many early mistakes.

## Updating `uv`

Occasionally:

```
uv self update
```

#### Tip

If it ever errors, you can just reinstall using the same one-liner from earlier.

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## Best practices for this course

- One folder for the course keeps everything tidy.
- Never install packages “globally” outside a project.
- Keep a short personal log in each project’s `README.md` (What did I do? What still confuses me?).
- Ask early for help, guessing usually takes much more time than asking.

You can always see available commands:

```
uv --help
```

## Recap

You can now:

1. Install `uv`.
2. Create a project.
3. Run a script.
4. Add/remove/sync packages.

Now, you’re set to continue the course.

## Bibliography